

WEB PERFORMANCE INITIATIVE · JUNE 2026

Sinister Diesel

Performance Improvement Plan

sinisterdiesel.com · Full GTmetrix Audit · HAR Analysis · CrUX Real-User Data

⚠ CORE WEB VITALS: FAILED (REAL USER DATA)

GTMETRIX GRADE

D

PERFORMANCE

53%

STRUCTURE

82%

LCP (LAB)

2.2s

TBT

407ms

CLS (REAL)

0.27

FULLY LOADED

7.9s

PAGE SIZE

6.07MB

REQUESTS

143

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Date: June 17, 2026 · Tests: 2 GTmetrix runs + HAR waterfall analysis

Tool: GTmetrix · Chrome 142 · Lighthouse 12.6.1 · Seattle, WA, USA

Platform: Miva Merchant · CDN: Cloudflare (HTTP/3 + Brotli active)

Executive Summary

Platform identified: Miva Merchant e-commerce · **CDN:** Cloudflare (already active — HTTP/3 & Brotli compression confirmed) · **Cache status:** **x-miva-cache: off** — page caching is currently disabled in store settings

Two GTmetrix runs were conducted on **June 17, 2026**, supplemented by a full HAR (network waterfall) export and Google CrUX real-user data. The site earns a **Grade D / 53% Performance** and takes **7.9 seconds** to fully load, transferring **6.07 MB** across **143 requests** from **35 third-party domains**.

The most critical finding is that **Google's Core Web Vitals assessment is currently FAILING** based on real user data — specifically due to Cumulative Layout Shift (CLS = 0.27) and Time to First Byte (TTFB = 1.2s). This directly impacts Google search rankings for sinisterdiesel.com today.

FULLY LOADED

7.9s

Target: under 3s

PAGE WEIGHT

6.07 MB

Images: 3.72MB · JS: 1.76MB

HTTP REQUESTS

143

68 JS · 33 images · 9 CSS · 8 fonts

THIRD-PARTY DOMAINS

35

Analytics, ads, loyalty, search, reviews

TBT (LAB)

407ms

Poor (>300ms threshold)

SPEED INDEX

5.7s

Very poor — visual load is slow

SEO Google uses Core Web Vitals as a search ranking signal — a failing CWV assessment means the site is being penalized in organic results today. **REVENUE** A 1-second improvement in load time can increase e-commerce conversions by 7%. At 7.9s, significant revenue is being lost to bounce. **GOOD NEWS** Cloudflare, HTTP/3, and Brotli are already deployed. The infrastructure is in place — configuration and content optimization are the gap.

Real-User Data & Detailed Timings

CrUX Core Web Vitals (Last 28 Days — Real Chrome Users)

These metrics reflect actual visitor experiences collected by Google. This data directly determines Google search ranking eligibility.

CLS · LAYOUT SHIFT

0.27

FAIL Target: < 0.1

TTFB · TIME TO FIRST BYTE

1.2s

NEEDS WORK Target: < 0.8s

LCP · LARGEST CONTENTFUL PAINT

1.7s

PASS Target: < 2.5s

INP · INTERACTION TO NEXT PAINT

79ms

PASS Target: < 200ms

FCP · FIRST CONTENTFUL PAINT

1.5s

PASS Target: < 1.8s

OVERALL CWV ASSESSMENT

FAILED

All 3 core vitals must pass

Lab Browser Timing Breakdown (GTmetrix Test 2)

TIMING MILESTONE	VALUE	WHAT IT MEANS	STATUS
Time to First Byte (TTFB)	1.1s	Server took 1.1s before sending any data	SLOW
First Paint	1.6s	First pixel drawn on screen	OK
First Contentful Paint (FCP)	1.6s	First text or image visible	NEEDS WORK
DOM Interactive	2.0s	HTML parsed; JS can run	SLOW
Largest Contentful Paint (LCP)	2.2s	Hero image/text fully visible	NEEDS WORK
Time to Interactive (TTI)	3.3s	Page responds to user input	POOR
DOM Content Loaded	2.3s	HTML + deferred scripts complete	SLOW
Onload Time	3.2s	All above-fold resources loaded	POOR
Fully Loaded Time	7.9s	Every resource on page complete	VERY POOR
Speed Index	5.7s	How fast page visually fills in	VERY POOR
Total Blocking Time (TBT)	407ms	JS blocking the main thread	POOR

CLS = 0.27 in Real Users (Lab Shows 0.1)

The lab test sees CLS of 0.1 but real Chrome users experience 0.27 — a FAIL. This gap suggests dynamic content (lazy-loaded images, ads, banners, Klaviyo pop-

Cloudflare Edge: 1,041ms

The HAR shows cfEdge;dur=1041 — Cloudflare spent 1,041ms processing the request at its edge before responding. With cf-cache-status: DYNAMIC, every page

ups) shifts layout after load for real users. This is the #1 priority fix for CWV compliance.

load hits the origin server. Enabling Cloudflare page caching alone could cut TTFB by ~800ms.

Root Cause Analysis — Specific Findings

Largest Individual Resources (from HAR waterfall)

SIZE	RESOURCE	TYPE	ISSUE	PRIORITY
971 KB	shopsocially-uploads.s3.amazonaws.com · social proof image	Image (PNG)	Third-party S3 image, uncompressed, no CDN edge	HIGH
752 KB	FathersDayDesktop.jpg	Image (JPEG)	Hero banner — not WebP, no lazy load, large viewport	HIGH
525 KB	GA FORD Desktop1.jpg	Image (JPEG)	Promo image — not WebP, cached only 1 day	HIGH
375 KB	www.gstatic.com · recaptcha__en.js	Script	Google reCAPTCHA loaded on every page, blocks thread	MED
189 KB	/72ai/ (inline AI/tracking script)	Script	Loaded synchronously on homepage, no defer	MED
174 KB	Screenshot 2025-02-21 090619_upscaled.jpg	Image	Filename suggests a raw screenshot used as web image	HIGH
150 KB	www.googletagmanager.com · gtm.js	Script	GTM blocks render; 35 tags likely firing on load	MED
145 KB	cdn1.affirm.com · affirm.js	Script	Buy-now-pay-later widget, loads on every page	MED
136 KB	connect.facebook.net · Facebook Pixel	Script	Fires immediately, adds latency for all users	LOW
118 KB	d1hmb9v9cg448y.cloudfront.net · Zinrelo loyalty JS	Script	Loyalty program widget loaded synchronously	MED

Cache Policy Audit — Images Are Only Cached 1 Day

ASSET TYPE	CURRENT CACHE	RECOMMENDED	SAVINGS
Homepage images (JPEG/PNG on mm5/)	max-age=86400 (1 day)	max-age=31536000 (1 year)	~2.83 MB per returning visitor
JavaScript files on sinisterdiesel.com	max-age=86400 (1 day)	max-age=31536000 (1 year)	Eliminates re-download for most users
Miva CSS/JS bundles (versioned)	max-age=31536000 ✓	Already optimal	—
SearchSpring bundle.js	max-age=1800 (30 min)	Third-party — request via vendor	Minor

ASSET TYPE	CURRENT CACHE	RECOMMENDED	SAVINGS
HTML pages	DYNAMIC (no cache)	Cloudflare cache rule for homepage	Biggest TTFB win

Third-Party Script Inventory (35 External Domains)

Heavy Hitters (Audit First)

- Google reCAPTCHA — 375 KB, every page
- Google Tag Manager — 150 KB, fires 35+ tags
- Affirm (BNPL) — 145 KB, can be deferred
- Facebook Pixel — 136 KB, can be deferred
- Zinrelo Loyalty — 118 KB, can be deferred
- CrazyEgg heatmaps — loads on every page

Also Present

- Klaviyo (email) — 4 separate scripts
- SearchSpring (search) — 2 scripts
- ShopperApproved (reviews)
- ShopSocially (social proof) — 971 KB image!
- Bing Ads pixel
- DoubleClick / Google Ads
- Cloudflare Insights beacon

12-Week Improvement Roadmap

Phases are ordered by **impact-per-effort ratio**. Phase 1 alone is estimated to bring the GTmetrix grade from D to C and move CWV closer to passing.

Phase 1 • Critical Quick Wins

WEEK 1-2 • LOW EFFORT • HIGHEST IMPACT • ZERO RISK

- **Fix CLS (Priority #1 — CWV FAIL):** Add explicit width and height to all `<img>` tags. Reserve space for dynamic content (ads, Klaviyo banners, ShopSocially widget) using CSS `min-height` or `aspect-ratio` containers. Target: CLS from 0.27 → under 0.1.
- **Compress & Convert Hero Images:** Convert `FathersDayDesktop.jpg` (752 KB) and `GA_FORD_Desktop1.jpg` (525 KB) to WebP. Estimated combined savings: ~850 KB. Use Squoosh or Cloudflare Images.
- **Fix "Screenshot" Image:** `Screenshot_2025-02-21_090619_upscaled.jpg` (174 KB) is a raw screenshot used as a web graphic. Replace with a properly exported, optimized asset.
- **Extend Image Cache Headers:** Change all `/mm5/graphics/` images from `max-age=86400` (1 day) to `max-age=31536000` (1 year). Saves 2.83 MB for every returning visitor.
- **Add Lazy Loading:** Add `loading="lazy"` to all below-the-fold images, especially category thumbnails and product photos.

Phase 2 • Server Response & Cloudflare Caching

WEEK 3-4 • MEDIUM EFFORT • BIGGEST TTFB GAIN

- **Enable Cloudflare Page Caching:** HAR shows `cf-cache-status: DYNAMIC` — Cloudflare is passing every request to origin. Create a Cloudflare Cache Rule to cache the homepage and category pages. Also investigate `x-miva-cache: off` store-settings in Miva admin to understand why Miva's own caching is disabled. Estimated impact: TTFB drops from 1.2s to under 200ms.
- **Enable Miva Page Caching:** Review Miva Merchant store settings for built-in caching options. Miva has native page fragment caching that should be enabled for non-personalised page sections.
- **Defer GTM Until After First Interaction:** Load Google Tag Manager after `DOMContentLoaded` or on first user interaction. This alone can reduce TBT by 50–100ms and removes a render-blocking 150 KB script from the critical path.
- **Serve Responsive Images:** Use `srcset` and `<picture>` so mobile users receive appropriately sized images instead of 752 KB desktop banners.

Phase 3 • JavaScript Audit & Third-Party Deferral

WEEK 5-8 • MEDIUM-HIGH EFFORT • TARGETS TBT + SPEED INDEX

- **reCAPTCHA (375 KB):** Move reCAPTCHA to render only on forms (checkout, contact). Do not load it on the homepage. Use reCAPTCHA v3 lazy injection.
- **Affirm BNPL (145 KB):** Defer Affirm JS until the product page or checkout stage. Do not load on homepage.
- **Zinrelo Loyalty (118 KB):** Defer until user scrolls or interacts with loyalty widget. Use `IntersectionObserver` to trigger load.
- **CrazyEgg Heatmaps:** Load asynchronously and only when not in a Lighthouse/bot context. Move to GTM as a tag to control firing.
- **Klaviyo Scripts:** Consolidate 4 separate Klaviyo script loads into one. Use Klaviyo's `async` embed method.
- **Audit GTM Tags:** Review all 35+ domains firing through GTM. Remove any tags for tools no longer in use. Pause non-essential tracking during off-peak seasons.

Phase 4 · DOM Cleanup & Ongoing Monitoring

WEEK 9-12 · ONGOING · PREVENTS REGRESSION

- **DOM Reduction:** Current DOM has 1,667 elements (recommended max: 800). Audit and remove redundant wrapper divs. Consider virtual scrolling for long product listing pages.
- **ShopSocially Image (971 KB):** The largest single resource is a third-party S3 PNG. Work with the ShopSocially vendor to serve a compressed WebP version, or proxy it through Cloudflare with image optimization rules.
- **GTmetrix Scheduled Monitoring:** Set up weekly automated GTmetrix tests so regressions are caught before they go live.
- **Google Search Console CWW Monitoring:** Set up alerts in Google Search Console for Core Web Vitals regressions. This shows real-user data from Google's index.
- **Performance Budget:** Define team-agreed limits (e.g., max page weight 3 MB, max 90 requests, LCP under 2.5s) to prevent future performance debt.

Target KPIs & Recommended First Tasks

Before / After Targets by Phase

Metric	Current	After Phase 1	After Phase 2-3	Final Goal
GTmetrix Grade	D	C	B	A
Performance Score	53%	65%	80%	90%+
CLS (Real Users)	0.27 X FAIL	0.08 ✓ PASS	< 0.05	< 0.1
TTFB (Real Users)	1.2s	1.0s	0.4s	< 0.8s
CWV Assessment	FAILED	Improving	PASSED	PASSED
Fully Loaded Time	7.9s	5.5s	3.5s	< 3.0s
Page Size	6.07 MB	3.5 MB	2.0 MB	< 2.0 MB
TBT	407ms	350ms	150ms	< 150ms
LCP (lab)	2.2s	1.8s	1.5s	< 1.5s
Speed Index	5.7s	4.0s	2.5s	< 2.5s

Recommended First Tasks for My Initial Sprint

#	Task	Expected Gain	Effort
1	Add width/height to all tags — especially for ads, Klaviyo banners, and ShopSocially widget. Reserve space with CSS to prevent CLS.	CLS: 0.27 → <0.1 (CWV PASS)	1-2 DAYS
2	Convert FathersDayDesktop.jpg + GA FORD Desktop1.jpg to WebP using Squoosh or Cloudflare Image Resizing. These 2 images alone = 1.28 MB.	~850 KB page size reduction	HALF DAY
3	Replace "Screenshot 2025-02-21" image — this 174 KB file appears to be a raw desktop screenshot used as a web graphic. Export a proper asset.	~150 KB savings + professionalism	1 HOUR
4	Extend image cache headers on /mm5/graphics/ from 1 day to 1 year. Verify with .htaccess or Cloudflare cache rule.	2.83 MB saved for return visitors	HALF DAY
5	Investigate Miva cache settings — x-miva-cache: off indicates store-level caching is disabled. Understand why and determine if it can be enabled safely.	Potential TTFB: 1.2s → <0.4s	RESEARCH
6	Run GTmetrix after each change to track delta and document before/after screenshots for reporting.	Measurable proof of progress	ONGOING

sinisterdiesel.com is currently **failing Google's Core Web Vitals** assessment, which impacts SEO rankings. The root causes are clear and specific: a 971 KB third-party image, two unoptimized hero banners (1.28 MB combined), 35 third-party script domains, and Miva/Cloudflare page caching that is currently turned off. The good news is that the infrastructure (Cloudflare, HTTP/3, Brotli) is already in place — this is a configuration and content problem, not an architecture rebuild. A focused 12-week plan starting with image optimization and CLS fixes (Week 1–2) can move the site to a CWV pass and GTmetrix Grade B, with Grade A achievable by Week 12.